

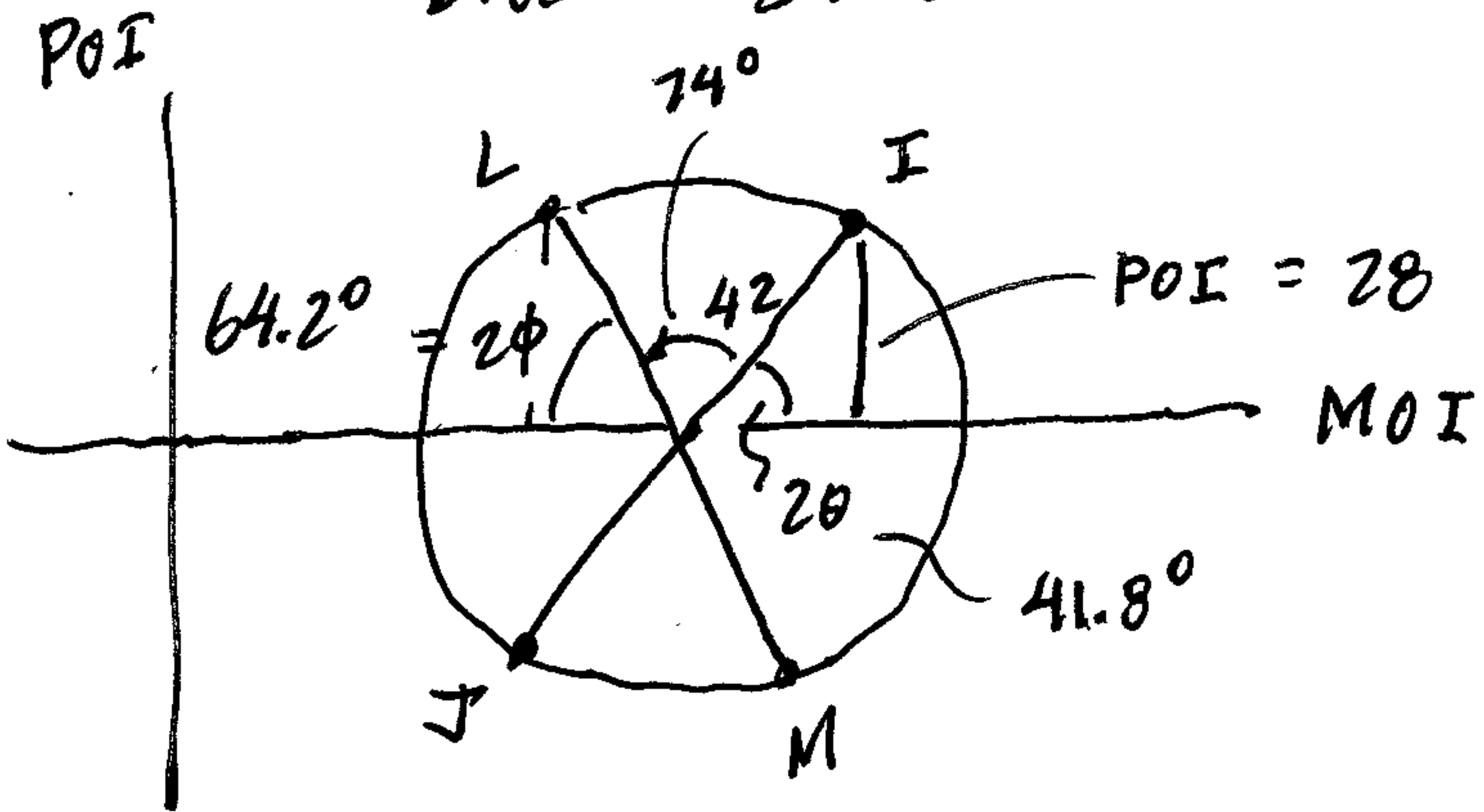
Another Exam 4 Practice Conceptual MC Problem

If the diameter of Mohr's Circle is 84 cm^4 and the product of inertia about a set of I-J axes is 28 cm^4 , determine the magnitude of the product of inertia about a set of L-M axes rotated 37° CCW on the cross section.

Solution

"Cross Section" is the original area of the beam shape.

$$Dia = 84 \text{ cm}^4 \Rightarrow R = 42 \text{ cm}^4$$



$$2\theta = \sin^{-1} \frac{28}{42}$$
$$= 41.8^\circ$$

37° on the "cross section"
 $= 2(37) = 74^\circ$ on MC

$$2\phi = 180 - 74 - 41.8$$
$$= 64.2^\circ$$

$$\begin{aligned} & \text{PoI at L} \\ &= R \cdot \sin 64.2^\circ \\ &= 42 \cdot \sin 64.2 \\ &= \boxed{37.8 \text{ cm} = \text{PoI at L}} \end{aligned}$$